INCH-POUND

MIL-DTL-83725/24A
17 October 2022
SUPERSEDING
MIL-R-83725/24(USAF)
23 November 1987

DETAIL SPECIFICATION SHEET

RELAYS, VACUUM, SPST, LATCHING, 15 AMPERES DC OR 60 HZ RMS, 5 KILOVOLTS PEAK

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the relays described herein shall consist of this specification and the latest issue of MIL-DTL-83725.

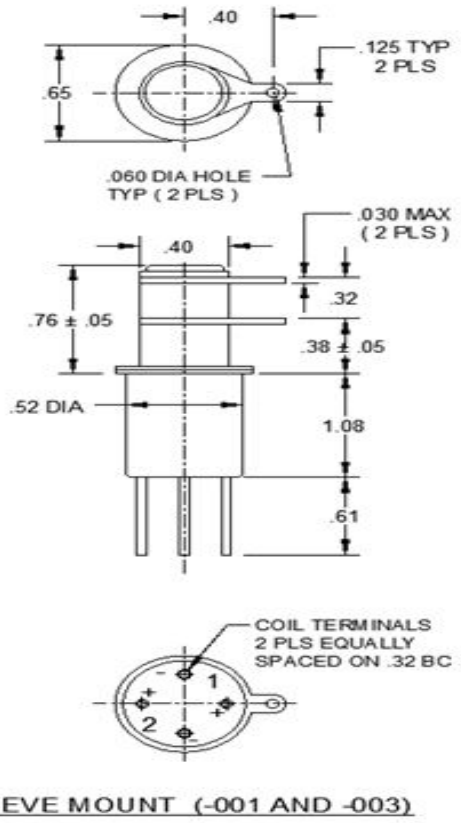
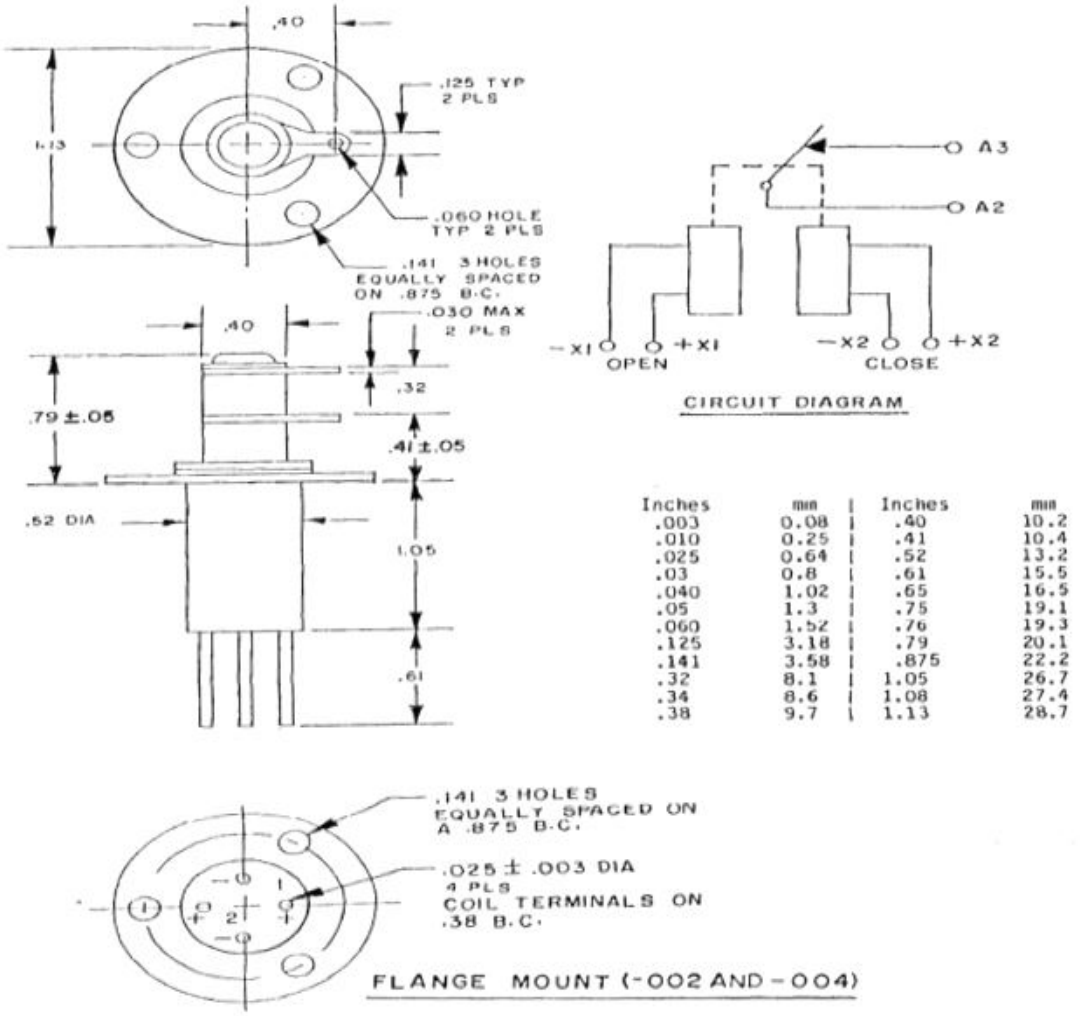


FIGURE 1. Relay dimensions and configuration.

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Inches	mm	Inches	mm
.003	0.08	.40	10.2
.010	0.25	.41	10.4
.025	0.64	.52	13.2
.03	0.8	.61	15.5
.040	1.02	.65	16.5
.05	1.3	.75	19.1
.060	1.52	.76	19.3
.125	3.18	.79	20.1
.141	3.58	.875	22.2
.32	8.1	1.05	26.7
.34	8.6	1.08	27.4
.38	9.7	1.13	28.7

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerances are ±.03 (0.76 mm) for two place decimals and ±.010 (0.25 mm) for three place decimals.
4. Glazing of ceramic finish is required.

FIGURE 1. Relay dimensions and configuration - Continued.

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REQUIREMENTS:

CONTACT DATA:

Configuration: SPDT, NC, ground isolated.
Arrangement: Latching.
Load ratings:
Resistive: (Carry only).
15 amperes dc or 60 Hz rms.
12 amperes rms, 2.5 Hz.
8 amperes rms, 16 MHz.
6 amperes rms, 32 MHz.
2.75 amperes rms, 50 MHz.
2.0 amperes rms, 75 MHz.

Rated operating voltage:
5 kilovolts peak, 60 Hz or dc.
4.5 kilovolts peak, 2.5 MHz.
3.5 kilovolts peak, 16 MHz.
2.8 kilovolts peak, 32 MHz.
2.0 kilovolts peak, 50 Hz.
1.0 kilovolt peak, 75 MHz.

Contact resistance:
Rated Life:
Before: .020 ohm, maximum.
During: .100 ohm, maximum.
After: .100 ohm, maximum.

Capacitance:
1.6 picofarads, maximum across open contacts.
1.6 picofarads, maximum between open contract and ground.

COIL DATA: 1/

Duty rating: 10% maximum duty cycle, 1 second maximum on time.
Operate time: -001 and -002, 4 milliseconds maximum, including bounce.
-003 and -004, 1 millisecond maximum, including bounce.
Pulse time: -001 and -004, 4 milliseconds maximum.
-003 and -004, 1 millisecond maximum.

ELECTRICAL DATA:

Insulation resistance: 1,000 megohms, minimum, except the resistance between coil and case at high temperature shall be 500 megohms or greater.

Dielectric withstanding voltage:

At atmospheric pressure, 60 Hz or dc.

Between all mated contacts in the open position:	6 kV peak.
Between high voltage terminals and case:	6 kV peak.
Between coil and case:	500 V rms, 60 Hz.

1/ CAUTION: The use of any coil voltage other than rated coil voltage may compromise the operation of the relay.

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ENVIRONMENTAL DATA:

Temperature range: -55°C to +125°C.
Vibration: MIL-STD-202-204, test condition C (10-55 Hz, .06 inch DA/55-2000 Hz, 10 g).
Shock: MIL-STD-202-213, test condition A, 50 g, 11 milliseconds, half sine.

PHYSICAL:

Terminal strength: 5 pounds pull.
Dimensions and configuration: See figure 1.
Termination: Solder terminal.
Weight: 1 ounce, maximum.

LIFE TEST REQUIREMENTS:

Mechanical cycling: 1,000,000 cycles. 2 sample units (cycle rate, 36,000 per hour maximum; contact current shall not exceed 10 milliamperes).

QUALITY ASSURANCE:

Dielectric withstanding voltage: Tests to be conducted at atmospheric pressure rating only.
Duration of application: 5-10 seconds at a 10 percent increase in the dielectric withstanding voltage.

PART NUMBER: M83725/24-001 and -003 sleeve mount or M83725/24-002 and -004 flange mount (see figure 1).

QUALIFICATION INSPECTION:

Qualification inspection: See table II.

TABLE II. Qualification inspection.

<table>
<tr><td rowspan="2">Single submission</td><td colspan="3">Group submission</td></tr>
<tr><td>Part</td><td>Sample size</td><td>Inspections</td></tr>
<tr><td rowspan="4">10 units plus
1 open unit</td><td>M83725/24-005</td><td>5 units</td><td rowspan="2">Full qualification</td></tr>
<tr><td>M83725/24-004</td><td>5 units</td></tr>
<tr><td>M83725/24-001</td><td>2 units</td><td rowspan="2">Group I of qualification inspection table, shock, vibration, plus 1 open unit of any dash number</td></tr>
<tr><td>M83725/24-002</td><td>2 units</td></tr>
</table>

LOGISTICS SUPPORT: The Government shall support M83725/24-001 with M83725/24-003 and M83725/24-002 with M83725/24-004.

Changes from previous issue. The margins of this specification are marked with vertical lines to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the previous issue.

Referenced documents. In addition to MIL-DTL-83725, this document references the following:
MIL-STD-202-204 MIL-STD-202-213

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Custodians:
Air Force – 85
DLA - CC

Preparing activity:
DLA-CC

(Project 5945-2022-051)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil/.

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